

Non-Contrast CT Segmentation

Evidence Summary — IngeniumAI AI Hackathon 2026

This document captures the key visual evidence gathered while testing IngeniumAI's baseline segmentation model (trained on contrast CT) directly on non-contrast CT scans, and after an initial fine-tuning attempt on a small set of real non-contrast image/label pairs.

1. Baseline model on non-contrast CT

The pretrained baseline model was run directly on a non-contrast CT scan with no adaptation. The pulmonary artery (yellow) is detected reasonably well, but the aorta (teal) is missed entirely.

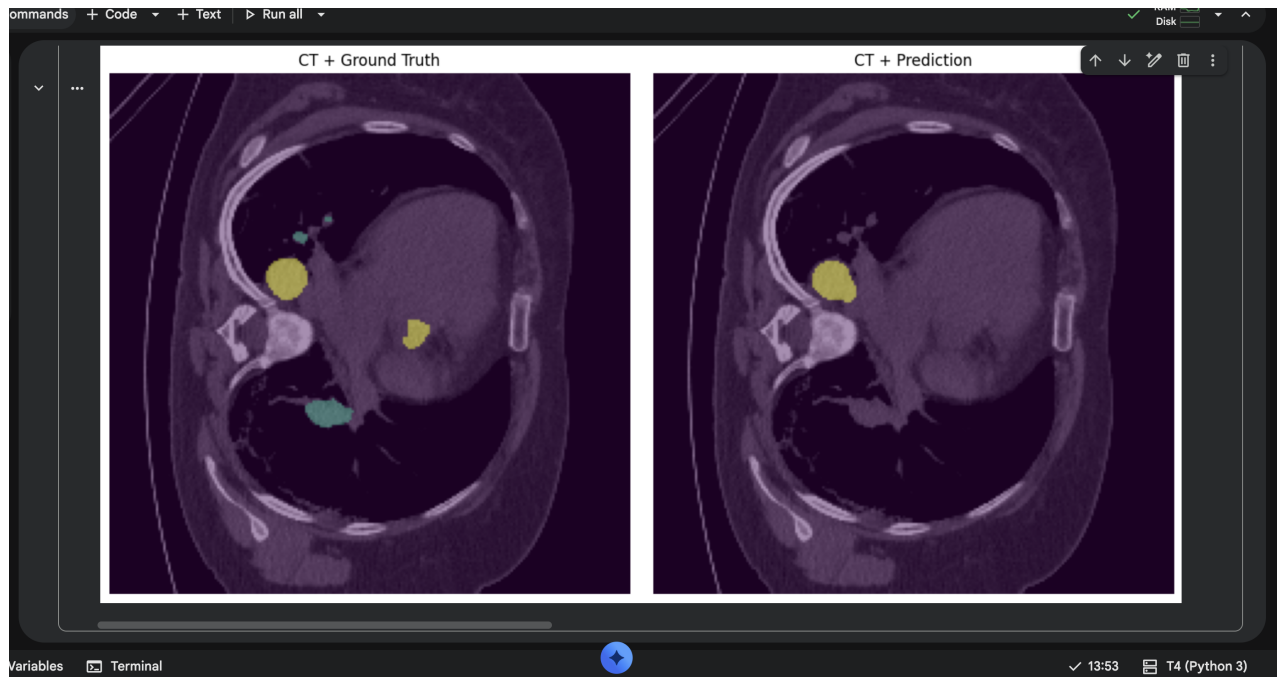


Figure 1. Left: CT slice with ground-truth segmentation overlay. Right: CT slice with the baseline model's prediction overlay. The pulmonary artery is captured; the aorta is absent from the prediction.

2. Baseline vs. fine-tuned comparison

The baseline model was fine-tuned on a small subset of real non-contrast CT image/label pairs. The panel below compares ground truth, the original baseline prediction, and the fine-tuned prediction on the same slice.

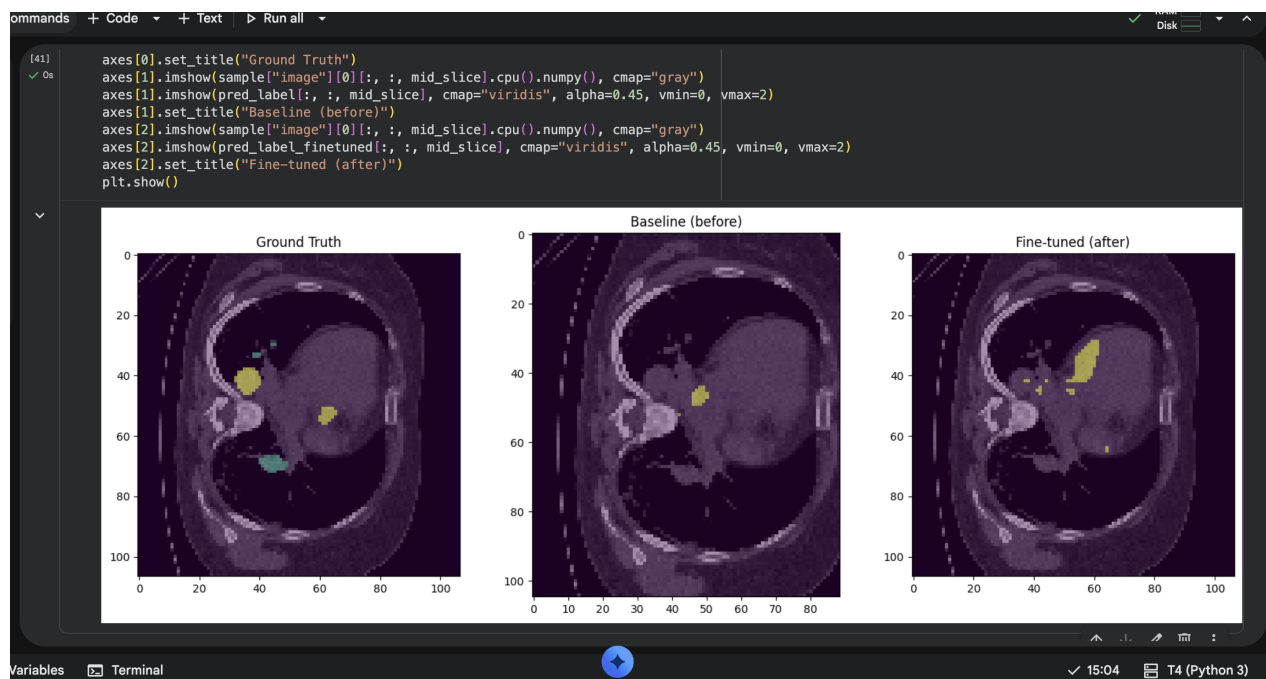


Figure 2. Left: ground truth. Middle: baseline prediction (pre-fine-tuning). Right: fine-tuned prediction. This early fine-tuning pass used a small training subset and few epochs as a feasibility check; the aorta remains undetected at this stage, indicating more training data and/or epochs are needed to close the gap fully.

Key takeaways

- The baseline model transfers partially to non-contrast CT: pulmonary artery detection holds up, likely due to consistent shape and anatomical position.
- Aorta detection fails completely at baseline, likely because it relies on contrast brightness that is absent in non-contrast scans, and its smooth round shape blends with surrounding structures.
- A minimal fine-tuning pass (small data subset, few epochs) was not sufficient to close this gap, suggesting the fix requires more training data, more epochs, and/or loss weighting toward the aorta class specifically.